

TIMOTHY WONG

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EXPERIENCE

Charles Schwab

Software Engineer II

May 2024 – Present

- Built SwarmPilot, a .NET/Avalonia application enabling the engineering team to orchestrate parallel GitHub Copilot agents across 24 repositories and automate CI/CD and Git workflows, reducing manual engineering effort by 90% while improving code consistency and accelerating debugging.
- Modernized authentication for a client-facing application, migrating credential management from legacy databases to a centralized authentication platform, redesigning login flows, and implementing MFA.
- Built a Kafka consumer to process authentication and credential events, improving observability and reducing production issue diagnosis time by more than 75%.
- Migrated the application's entire frontend from legacy ASP.NET MVC to Angular, building reusable components and integrating backend REST APIs.
- Replaced legacy integrations with new REST and orchestration APIs to support third-party partner onboarding.

Software Engineer I

Sep 2021 – May 2024

- Developed Angular components and .NET Web API endpoints for participant and administrator retirement applications, supporting new business workflows and service integrations.
- Investigated and resolved production issues across client facing applications to maintain stability.
- Expanded automated regression coverage using Java and Gherkin, reducing manual testing effort.

PROJECTS

[AI Infrastructure Research Terminal](#) | TypeScript, Next.js, PostgreSQL, Redis, BullMQ

- Developed a research platform that ingests SEC filings and investor materials to generate source-cited answers and company comparisons, with background processing through Redis and BullMQ.
- Integrated PostgreSQL full text search and pgvector similarity search with analyst approved evidence, citation and numeric validation, and saved source snapshots to make generated research auditable.

[GradientGuard](#) | TypeScript, React, Canvas API, Vitest, Playwright

- Built a gradient editor that samples canvas pixels across text regions to detect and visualize contrast issues.
- Designed a binary search optimizer to calculate the minimum overlay opacity needed to meet contrast targets across sampled points and added CSS export and URL based design sharing.

TECHNICAL SKILLS

Languages – TypeScript, Python, Java, C#, JavaScript, SQL

Frontend – React, Next.js, Angular, Tailwind CSS

Backend – Node.js, .NET, REST APIs

Databases – PostgreSQL, Redis, SQL Server, Oracle

Infrastructure – Kafka, Docker, Vercel

Tools – Git, GitHub, Splunk, Postman, Figma, Codex, GitHub Copilot

EDUCATION

University of Maryland, College Park, MD

May 2021

Bachelor of Science, Computer Science & Economics